

**СИСТЕМА ПРОСТОРОВОГО АНАЛІЗУ ГРИ В ВОЛЕЙБОЛ З
ВИКОРИСТАННЯМ МАШИННОГО НАВЧАННЯ**

Текст програми

ІА/Ц.467200.007 Д4

ПЕРЕЛІК ФАЙЛІВ ПРОГРАМНОГО КОМПЛЕКСУ

У цьому додатку наведено повні вихідні тексти ключових модулів програмного комплексу системи трикутного просторового аналізу польоту волейбольного м'яча. Модулі впорядковано відповідно до архітектурного графа залежностей, описаного в розділі 3 пояснювальної записки: спочатку наводиться модуль геометричного зворотного проектування, далі — ядро байєсової фільтрації, потім модуль трекінгу зі скінченням автоматом життєвого циклу треку та, нарешті, сценарій навчання згорткової нейромережі-детектора. Сумарний обсяг наведеного коду становить понад 950 рядків мовою програмування Python.

ball_3d_analysis/get_court_position_methods.py

```
import cv2
import numpy as np
import torch as t
import torchvision
import torchaudio
import albumentations as A
from ultralytics import YOLO
```

Reference calibration correspondences: the four court corners.

Court is 9 m wide x 18 m long; world coordinates are in metres.

```
pts_real_3d = np.array([
    [0.0, 0.0, 0.0],
```

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Перевірив	Морозов-Леонів							Аркуші
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Н. контр.	Міщенко Л. Д.						НТУУ "КПІ ім. Ігоря Сікорського", ФІОТ ІО-21	
Затвердив	Волокита А. М.							

```
[9.0, 0.0, 0.0],
[9.0, 0.0, 18.0],
[0.0, 0.0, 18.0]],dtype=np.float32)
```

Pixel locations of the same four corners in the broadcast frame.

```
pts_video_2d = np.array([
    [ 69, 1033], # P0 front-left
    [1840, 1031], # P1 front-right
    [1507, 609], # P2 back-right
    [ 405, 609]], # P3 back-left
    dtype=np.float32)
```

Initial pinhole intrinsics (principal point at the image centre). The focal
length is only a starting guess here — it is refined by calibration.

```
K = np.array([
    [1300.0, 0.0, 960.0],
    [0.0, 1300.0, 540.0],
    [0.0, 0.0, 1.0]],dtype=np.float32)
```

```
def calibrate_camera(pts_3d, pts_2d, camera_matrix, dist, up_axis=1):
```

```
    """
```

Calibrate the camera with PnP. Returns (R, tvec, camera_position):

R — world->camera rotation matrix (3x3),

tvec — translation into the camera frame (3,1),

camera_position — camera location in world coordinates (3,1).

up_axis flips R so the camera ends up above the floor: the four coplanar

court corners leave the vertical direction sign-ambiguous otherwise.

''''''

```
success, rvec, tvec = cv2.solvePnP(pts_3d, pts_2d, camera_matrix, dist,
flags=cv2.SOLVEPNP_ITERATIVE)
```

```
if not success:
```

```
    raise ValueError("solvePnP failed to converge; "
        "check the order of the point correspondences")
```

```
R, _ = cv2.Rodrigues(rvec)
```

```
# Camera position in world coordinates (C = -R^T * tvec).
```

```
R_inv = np.linalg.inv(R)
```

```
camera_position = -np.dot(R_inv, tvec)
```

```
if (up_axis is not None
```

```
    and 0 <= up_axis < 3
```

```
    and camera_position[up_axis, 0] < 0.0):
```

```
    flip = np.eye(3)
```

```
    flip[up_axis, up_axis] = -1.0
```

```
    R = np.dot(R, flip)
```

```
    camera_position = np.dot(flip, camera_position)
```

```
return R, tvec, camera_position
```

```
def get_3d_position(u, v, bbox_w, K, R_matrix, camera_pos,
```

```
    ball_diameter=0.21, z_min=2.0, z_max=25.0,
```

```
    return_none_on_clip=False):
```

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''''''

Back-project a ball detection into a 3D world position.

Args:

u, v: ball bbox centre, in pixels.

bbox_w: ball bbox width, in pixels. Width is used (not height)
because it suffers less from motion blur.

K: camera intrinsics matrix.

R_matrix: rotation matrix from solvePnP.

camera_pos: 3D camera position (C) from solvePnP.

ball_diameter: real volleyball diameter, in metres (21 cm nominal).

z_min, z_max: sanity bounds on Z_c (distance from camera along the ray).

Returns:

np.ndarray (3,) — world position [X, Y, Z]; or None when

return_none_on_clip=True and Z_c fell outside [z_min, z_max].

''''''

f_x = K[0, 0]

c_x = K[0, 2]

f_y = K[1, 1]

c_y = K[1, 2]

Z_c = (f_x * ball_diameter) / bbox_w

if Z_c < z_min or Z_c > z_max:

if return_none_on_clip:

return None

Z_c = max(z_min, min(z_max, Z_c))

```

x_norm = (u - c_x) / f_x
y_norm = (v - c_y) / f_y
d_cam = np.array([[x_norm], [y_norm], [1.0]])

```

```

d_world = np.dot(R_matrix.T, d_cam)
P_world = camera_pos.reshape(3, 1) + Z_c * d_world

return P_world.ravel()

```

```

def get_3d_position_with_cov(u, v, bbox_w, K, R_matrix, camera_pos,
    ball_diameter=0.21, sigma_w=1.0, sigma_uv=1.0, r_floor=0.02,
    z_min=2.0, z_max=25.0, return_none_on_clip=False):
    """Like get_3d_position, but also returns a 3x3 world-frame measurement
    covariance R, anisotropic and consistent with monocular depth geometry.

```

Depth error follows from $Z_c = f \cdot D / w_{\text{box}}$ linearised in the bbox width:

$$\sigma_{Z_c} = |dZ_c/dw| \cdot \sigma_w = (f \cdot D / w_{\text{box}}^2) \cdot \sigma_w,$$

so depth is trusted less for distant (small- w) balls. The uncertainty lies along the ray, giving a rank-1 ellipsoid plus lateral pixel terms:

$$\text{Cov} \sim \sigma_{Z_c}^2 \cdot d \cdot d^T + Z_c^2 \cdot \sigma_{uv}^2 \cdot (J_u J_u^T + J_v J_v^T).$$

Args:

σ_w : bbox-width sigma, px (detector noise). Default 1.0.

σ_{uv} : bbox-centre sigma in u,v, px. Default 1.0.

r_{floor} : isotropic variance floor (m^2) on every axis. Default 0.02

(lateral term of the old fixed R), keeping lateral trust stable and

R positive-definite; only the along-ray depth stays adaptive.

Returns:

(P_world (3,), R_cov (3,3)); or (None, None) on clip when
return_none_on_clip=True.

''''''

f_x = K[0, 0]

c_x = K[0, 2]

f_y = K[1, 1]

c_y = K[1, 2]

Z_c = (f_x * ball_diameter) / bbox_w

if Z_c < z_min or Z_c > z_max:

if return_none_on_clip:

return None, None

Z_c = max(z_min, min(z_max, Z_c))

x_norm = (u - c_x) / f_x

y_norm = (v - c_y) / f_y

d_cam = np.array([[x_norm], [y_norm], [1.0]])

d_world = np.dot(R_matrix.T, d_cam) # (3,1) ray direction

P_world = camera_pos.reshape(3, 1) + Z_c * d_world

Depth sigma along the ray, from the bbox-width error.

sigma_Zc = (f_x * ball_diameter) / (bbox_w ** 2) * sigma_w

Lateral Jacobians (point shift per 1 px move of the bbox centre).

```

j_u = np.dot(R_matrix.T, np.array([[1.0 / f_x], [0.0], [0.0]]))
j_v = np.dot(R_matrix.T, np.array([[0.0], [1.0 / f_y], [0.0]]))

R_cov = (sigma_Zc ** 2) * np.dot(d_world, d_world.T)
R_cov += (Z_c ** 2) * (sigma_uv ** 2) * np.dot(j_u, j_u.T)
R_cov += (Z_c ** 2) * (sigma_uv ** 2) * np.dot(j_v, j_v.T)

# Isotropic floor: keeps lateral trust at the proven level of the old R
# (0.02) and guarantees positive-definiteness. Only the along-ray depth
# stays adaptive (sigma_Zc^2 on top of the floor).
R_cov += np.eye(3) * r_floor

return P_world.ravel(), R_cov.astype(np.float64)

```

ball_3d_analysis/IMM_UKF.py

```

import numpy as np
from copy import deepcopy
from scipy.linalg import cholesky, inv, expm, block_diag
from get_court_position_methods import calibrate_camera, get_3d_position
from scipy.stats import multivariate_normal
import sys

```

```

class UnscentedKalmanFilter(object):

```

```

    def __init__(self, name, dim_x, dim_z, dt, fx, hx, points, sqrt_fn=None,
                  x_mean_fn=None, z_mean_fn=None, residual_x=None, residual_z=None):

```

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```

self.name = name
self.x = np.zeros(dim_x)
self.P = np.eye(dim_x)
self.x_prior = np.copy(self.x)
self.P_prior = np.copy(self.P)
self.Q = np.eye(dim_x)
self.R = np.eye(dim_z)
self._dim_x = dim_x
self._dim_z = dim_z
self.points_fn = points
self._dt = dt
self._num_sigmas = points.num_sigmas()
self.hx = hx
self.fx = fx
self.x_mean_fn = x_mean_fn
self.z_mean_fn = z_mean_fn
self.Wm, self.Wc = points.Wm, points.Wc
self.likelihood = None
self._log_likelihood = np.log(sys.float_info.min)
self._mahalanobis = None

if sqrt_fn is None:
    self.msqrt = cholesky
else:
    self.msqrt = sqrt_fn

if residual_x is None:
    self.residual_x = np.subtract

```

```

else:
    self.residual_x = residual_x

if residual_z is None:
    self.residual_z = np.subtract
else:
    self.residual_z = residual_z

# sigma points transformed through f(x) and h(x)
self.sigmas_f = np.zeros((self._num_sigmas, self._dim_x))
self.sigmas_h = np.zeros((self._num_sigmas, self._dim_z))

self.K = np.zeros((dim_x, dim_z))
self.y = np.zeros(dim_z)
self.z = np.array([[None]*dim_z])
self.S = np.zeros((dim_z, dim_z))
self.SI = np.zeros((dim_z, dim_z))
self.inv = np.linalg.inv

self.x_prior = self.x.copy()
self.P_prior = self.P.copy()

self.x_post = self.x.copy()
self.P_post = self.P.copy()

def predict(self, dt=None, UT=None, fx=None, **fxargs):
    """
    :param dt: time step for next iteration of computation

```

:param UT: unscented transform function

:param fx: state transition function

:param fxargs: arguments passed to f(x)

''''''

if dt is None:

dt = self._dt

if UT is None:

UT = unscented_transform

calculating sigma points for given mean and covariance

self.compute_process_sigmas(dt, fx, **fxargs)

#pass sigmas through the unscented transform to compute the prior

self.x, self.P = UT(self.sigmas_f, self.Wm, self.Wc,

self.Q, self.x_mean_fn, self.residual_x)

self.P = (self.P + self.P.T) / 2.0

self.x_prior = np.copy(self.x)

self.P_prior = np.copy(self.P)

def update(self, z, R=None, UT=None, hx=None, **hxargs):

''''''

:param z: measurement vector

:param R: measurement noise matrix

:param UT: function for unscented transform

:param hx: measurement function

:param hxargs: arguments for hx

''''''

if z is None:

```

self.z = np.array([[None]*self._dim_z]).T
self.x_post = self.x.copy()
self.P_post = self.P.copy()
self.likelihood = 1.0
return

if hx is None:
    hx = self.hx

if UT is None:
    UT = unscented_transform

if R is None:
    R = self.R
elif np.isscalar(R):
    R = np.eye(self._dim_z) * R

# passing prior sigmas through h(x) to get measurement sigmas
# the shape of sigmas_h will vary if the shape of z varies
sigmas_h = []
for s in self.sigmas_f:
    sigmas_h.append(hx(s, **hxargs))

self.sigmas_h = np.atleast_2d(sigmas_h)

# mean and covariance of prediction passed through unscented transform
z_mean, self.S = UT(self.sigmas_h, self.Wm, self.Wc,
                    R, self.z_mean_fn, self.residual_z)

```

```

# Regularise S before inversion / likelihood: guarantees positive
# definiteness even when the covariance is momentarily degenerate.
S_stable = self.S + np.eye(self.S.shape[0]) * 1e-6
try:
    self.SI = np.linalg.inv(S_stable)
except np.linalg.LinAlgError:
    self.SI = np.linalg.pinv(S_stable)

# compute cross variance of the state and measurement
Pxz = self.cross_variance(self.x, z_mean, self.sigmas_f, self.sigmas_h)

self.K = np.dot(Pxz, self.SI) # Kalman gain
self.y = self.residual_z(z, z_mean)

try:
    self.likelihood = multivariate_normal.pdf(
        self.y,
        mean=np.zeros_like(self.y),
        cov=S_stable,
        allow_singular=True,
    )
except (np.linalg.LinAlgError, ValueError):
    self.likelihood = 1e-300

if not np.isfinite(self.likelihood) or self.likelihood <= 0.0:
    self.likelihood = 1e-300

# updating Gaussian state estimate(x, P)

```

```

self.x = self.x + np.dot(self.K, self.y)
self.P = self.P - np.dot(self.K, np.dot(self.S, self.K.T))

# save measurement and posterior state
self.z = deepcopy(z)
self.x_post = self.x.copy()
self.P_post = self.P.copy()

def cross_variance(self, x, z, sigmas_f, sigmas_h):
    # Computing cross variance of the state 'x' and measurement 'z'
    Pxz = np.zeros((sigmas_f.shape[1], sigmas_h.shape[1]))
    N = sigmas_f.shape[0]
    for i in range(N):
        dx = self.residual_x(sigmas_f[i], x)
        dz = self.residual_z(sigmas_h[i], z)
        Pxz += self.Wc[i] * np.outer(dx, dz)
    return Pxz

def compute_process_sigmas(self, dt, fx=None, **fx_args):
    # computes the values of sigmas_f
    if fx is None:
        fx = self.fx

    # calculate sigma points for given mean and covariance
    sigmas = self.points_fn.sigma_points(self.x, self.P)

    for i, s in enumerate(sigmas):
        self.sigmas_f[i] = fx(s, dt, **fx_args)

```

@property

def log_likelihood(self):

"""

log-likelihood of the last measurement.

"""

if self._log_likelihood is None:

self._log_likelihood = logpdf(x=self.y, cov=self.S)

return self._log_likelihood

@property

def mahalanobis(self):

"""

Mahalanobis distance of measurement. E.g. 3 means measurement was 3 standard deviations away from the predicted value.

"""

if self._mahalanobis is None:

self._mahalanobis = np.sqrt(float(np.dot(np.dot(self.y.T, self.SI), self.y)))

return self._mahalanobis

class IMMEstimator(object):

""" Implements an Interacting Multiple-Model (IMM) estimator.

:param filters: N-list consisting of filters in sequenced order

:param mu: (N) array-like of float with mode probabilities for each filter

:param M: (N, N) Markov chain transition matrix

"""

```

def __init__(self, filters, mu, M):
    if len(filters) < 2:
        raise ValueError('filters must contain at least two filters')

    self.filters = filters
    self.mu = np.asarray(mu) / np.sum(mu)
    self.M = M

    x_shape = filters[0].x.shape
    for f in filters:
        if x_shape != f.x.shape:
            raise ValueError(
                'All filters must have the same state dimension')

    self.x = np.zeros(filters[0].x.shape)
    self.P = np.zeros(filters[0].P.shape)
    self.N = len(filters)
    self.likelihood = np.zeros(self.N)
    self.omega = np.zeros((self.N, self.N))
    self._compute_mixing_probabilities()

    # initialize imm state estimate based on current filters
    self._compute_state_estimate()
    self.x_prior = self.x.copy()
    self.P_prior = self.P.copy()
    self.x_post = self.x.copy()
    self.P_post = self.P.copy()
    self._log_likelihood = np.log(sys.float_info.min)

```



```
self._mahalanobis = None
```

```
def filter_setx(self):
```

```
    for f in self.filters:
```

```
        f.x = self.x
```

```
def update(self, z, R=None):
```

```
    """
```

Add a new measurement (z) to the Kalman filter. If z is None, nothing is changed.

:param z: measurement

:param R: optional per-detection measurement covariance (3x3). If None, each filter uses its own fixed self.R. Passing a per-detection R (adaptive sigma_Z) makes depth trust depend on the bbox width.

```
    """
```

```
# run update on each filter, and save the likelihood
```

```
for i, f in enumerate(self.filters):
```

```
    f.update(z, R=R)
```

```
    self.likelihood[i] = f.likelihood
```

```
# update mode probabilities from total probability * likelihood
```

```
self.mu = self.cbar * self.likelihood
```

```
sum_mu = np.sum(self.mu)
```

```
if sum_mu == 0.0:
```

```
    self.mu = self.cbar.copy()
```

```
else:
```

```
    self.mu /= sum_mu
```

```

self._compute_mixing_probabilities()

# compute mixed IMM state and covariance and save posterior estimate
self._compute_state_estimate()
self.x_post = self.x.copy()
self.P_post = self.P.copy()

def predict(self, u=None):
    """
    Predict next state (prior) using the IMM state propagation
    equations.
    :param u: control vector, if None then none
    """

    # compute mixed initial conditions
    xs, Ps = [], []
    for i, (f, w) in enumerate(zip(self.filters, self.omega.T)):
        x = np.zeros(self.x.shape)
        for kf, wj in zip(self.filters, w):
            x += kf.x * wj
        xs.append(x)

        P = np.zeros(self.P.shape)
        for kf, wj in zip(self.filters, w):
            y = kf.x - x
            P += wj * (np.outer(y, y) + kf.P)
        Ps.append(P)

```

```

# compute each filter's prior using the mixed initial conditions
for i, f in enumerate(self.filters):
    # propagate using the mixed state estimate and covariance
    f.x = xs[i].copy()
    f.P = Ps[i].copy()
    f.predict(u)

# compute mixed IMM state and covariance and save posterior estimate
self._compute_state_estimate()
self.x_prior = self.x.copy()
self.P_prior = self.P.copy()

def _compute_state_estimate(self):
    """
    Computes the IMM's mixed state estimate from each filter using
    the the mode probability self.mu to weight the estimates.
    """
    self.x.fill(0)
    for f, mu in zip(self.filters, self.mu):
        self.x += f.x * mu

    self.P.fill(0)
    for f, mu in zip(self.filters, self.mu):
        y = f.x - self.x
        self.P += mu * (np.outer(y, y) + f.P)

def _compute_mixing_probabilities(self):

```

```
# Compute the mixing probability for each filter
```

```
self.cbar = np.dot(self.mu, self.M)
```

```
for i in range(self.N):
```

```
    for j in range(self.N):
```

```
        self.omega[i, j] = (self.M[i, j]*self.mu[i]) / self.cbar[j]
```

```
@property
```

```
def log_likelihood(self):
```

```
    """
```

```
    log-likelihood of the last measurement.
```

```
    """
```

```
    if self._log_likelihood is None:
```

```
        self._log_likelihood = logpdf(x=self.y, cov=self.S)
```

```
    return self._log_likelihood
```

```
@property
```

```
def mahalanobis(self):
```

```
    """
```

```
    Mahalanobis distance of measurement. E.g. 3 means measurement  
    was 3 standard deviations away from the predicted value.
```

```
    """
```

```
    if self._mahalanobis is None:
```

```
        self._mahalanobis = np.sqrt(float(np.dot(np.dot(self.y.T, self.SI), self.y)))
```

```
    return self._mahalanobis
```

class MerweScaledSigmaPoints(object):

''''''

Generates sigma points and weights according to Van der Merwe's 2004 dissertation[1] for the UnscentedKalmanFilter class. It parametrizes the sigma points using alpha, beta, kappa terms, and is the version seen in most publications.

:param n : int

Dimensionality of the state. $2n+1$ weights will be generated.

:param alpha : float

Determines the spread of the sigma points around the mean.

Usually a small positive value ($1e-3$).

:param beta : float

Incorporates prior knowledge of the distribution of the mean. For Gaussian α $\beta=2$ is optimal.

:param kappa : float, default=0.0

Secondary scaling parameter usually set to 0 or to $3-n$.

:param sqrt_method : function(ndarray), default=scipy.linalg.cholesky

Defines how we compute the square root of a matrix, which has no unique answer. Cholesky is the default choice due to its speed (another good choice - `scipy.linalg.sqrtm`).

:param subtract : callable (x, y), optional

Function that computes the difference between x and y.

You will have to supply this if your state variable cannot support subtraction, such as angles (359-1 degrees is 2, not 358). x and y are state vectors, not scalars.

Attributes:

Wm : np.array

weight for each sigma point for the mean

Wc : np.array

weight for each sigma point for the covariance

~~~~~

```
def __init__(self, n, alpha, beta, kappa, sqrt_method=None, subtract=None):
```

```
#pylint: disable=too-many-arguments
```

```
self.n = n
```

```
self.alpha = alpha
```

```
self.beta = beta
```

```
self.kappa = kappa
```

```
if sqrt_method is None:
```

```
    self.sqrt = cholesky
```

```
else:
```

```
    self.sqrt = sqrt_method
```

```
if subtract is None:
```

```
    self.subtract = np.subtract
```

```
else:
```

```
self.subtract = subtract
```

```
self._compute_weights()
```

```
def num_sigmas(self):
```

```
    """ Number of sigma points for each variable in the state x """
```

```
    return 2*self.n + 1
```

```
def sigma_points(self, x, P):
```

```
    """ Computes the sigma points for an unscented Kalman filter
```

```
    given the mean (x) and covariance(P) of the filter.
```

```
    Returns tuple of the sigma points and weights.
```

Works with both scalar and array inputs:

```
sigma_points (5, 9, 2) # mean 5, covariance 9
```

```
sigma_points ([5, 2], 9*eye(2), 2) # means 5 and 2, covariance 9I
```

```
:param x : An array-like object of the means of length n
```

```
:param P : scalar, or np.array
```

```
    Covariance of the filter. If scalar, is treated as eye(n)*P.
```

```
:return np.array, of size (n, 2n+1)
```

Two dimensional array of sigma points. Each column contains all of the sigmas for one dimension in the problem space.

```

Ordered by  $X_{i_0}$ ,  $X_{i_{\{1..n\}}}$ ,  $X_{i_{\{n+1..2n\}}}$ 
"""

if self.n != np.size(x):
    raise ValueError("expected size(x) {}, but size is {}".format(
        self.n, np.size(x)))

n = self.n

if np.isscalar(x):
    x = np.asarray([x])

if np.isscalar(P):
    P = np.eye(n)*P
else:
    P = np.atleast_2d(P)

lambda_ = self.alpha**2 * (n + self.kappa) - n
U = self.sqrt((lambda_ + n)*P)

sigmas = np.zeros((2*n+1, n))
sigmas[0] = x
for k in range(n):
    # pylint: disable=bad-whitespace
    sigmas[k+1] = self.subtract(x, -U[k])
    sigmas[n+k+1] = self.subtract(x, U[k])

return sigmas

```



```
def _compute_weights(self):
    """ Computes the weights for the scaled unscented Kalman filter. """

    n = self.n
    lambda_ = self.alpha**2 * (n + self.kappa) - n

    c = .5 / (n + lambda_)
    self.Wc = np.full(2*n + 1, c)
    self.Wm = np.full(2*n + 1, c)
    self.Wc[0] = lambda_ / (n + lambda_) + (1 - self.alpha**2 + self.beta)
    self.Wm[0] = lambda_ / (n + lambda_)
```

```
def __repr__(self):
    return (f'MerweScaledSigmaPoints(n={self.n}, alpha={self.alpha}, '
            f'beta={self.beta}, kappa={self.kappa})')
```

```
def unscented_transform(sigmas, Wm, Wc, noise_cov=None,
                        mean_fn=None, residual_fn=None):
```

```
    """
```

Computes unscented transform of a set of sigma points and weights.

returns the mean and covariance in a tuple.

:param residual\_fn:

:param sigmas: ndarray, of size (n, 2n+1)

2D array of sigma points.

:param Wm : ndarray [# sigmas per dimension]

Weights for the mean.

:param Wc : ndarray [# sigmas per dimension]

Weights for the covariance.

:param noise\_cov: ndarray, optional

noise matrix added to the final computed covariance matrix.

:param mean\_fn: callable (sigma\_points, weights), optional

Function that computes the mean of the provided sigma points  
and weights

''''''

kmax, n = sigmas.shape

try:

if mean\_fn is None:

# new mean is just the sum of the sigmas \* weight

x = np.dot(Wm, sigmas) # dot =  $\sum (W[k] * X_i[k])$

else:

x = mean\_fn(sigmas, Wm)

except:

print(sigmas)

raise

```
# new covariance is the sum of the outer product of the residuals
# times the weights
```

```
# this is the fast way to do this - see 'else' for the slow way
```

```
if residual_fn is np.subtract or residual_fn is None:
```

```
    y = sigmas - x[np.newaxis, :]
```

```
    P = np.dot(y.T, np.dot(np.diag(Wc), y))
```

```
else:
```

```
    P = np.zeros((n, n))
```

```
    for k in range(kmax):
```

```
        y = residual_fn(sigmas[k], x)
```

```
        P += Wc[k] * np.outer(y, y)
```

```
if noise_cov is not None:
```

```
    P += noise_cov
```

```
return x, P
```

```
def Q_discrete_white_noise(dim, dt=1., var=1., block_size=1, order_by_dim=True):
```

```
    """
```

Returns the Q matrix for the Discrete Constant White Noise

Model. dim may be either 2, 3, or 4 dt is the time step, and sigma

is the variance in the noise.

Q is computed as the  $G * G^T * \text{variance}$ , where G is the process noise per time step. In other words,  $G = [[.5dt^2][dt]]^T$  for the constant velocity

model.

:param dim : int (2, 3, or 4)

dimension for Q, where the final dimension is (dim x dim)

:param dt : float, default=1.0

time step in whatever units your filter is using for time. i.e. the amount of time between innovations

:param var : float, default=1.0

variance in the noise

:param block\_size : int >= 1

If your state variable contains more than one dimension, such as a 3d constant velocity model  $[x \ x' \ y \ y' \ z \ z']^T$ , then Q must be a block diagonal matrix.

:param order\_by\_dim : bool, default=True

Defines ordering of variables in the state vector. 'True' orders by keeping all derivatives of each dimensions)

$[x \ x' \ x'' \ y \ y' \ y'']$

whereas 'False' interleaves the dimensions

$[x \ y \ z \ x' \ y' \ z' \ x'' \ y'' \ z'']$

''''''

if not (dim == 2 or dim == 3 or dim == 4):

raise ValueError("dim must be between 2 and 4")

```

if dim == 2:
    Q = [[.25*dt**4, .5*dt**3],
          [.5*dt**3, dt**2]]
elif dim == 3:
    Q = [[.25*dt**4, .5*dt**3, .5*dt**2],
          [.5*dt**3, dt**2, dt],
          [.5*dt**2, dt, 1]]
else:
    Q = [(dt**6)/36, (dt**5)/12, (dt**4)/6, (dt**3)/6],
          [(dt**5)/12, (dt**4)/4, (dt**3)/2, (dt**2)/2],
          [(dt**4)/6, (dt**3)/2, dt**2, dt],
          [(dt**3)/6, (dt**2)/2, dt, 1.]]

if order_by_dim:
    return block_diag(*[Q]*block_size) * var
return order_by_derivative(np.array(Q), dim, block_size) * var

def logpdf(x, mean=None, cov=1, allow_singular=True):
    """
    Computes the log of the probability density function of the normal
    N(mean, cov) for the data x. The normal may be univariate or multivariate.
    """

    if mean is not None:
        flat_mean = np.asarray(mean).flatten()
    else:
        flat_mean = None

```

```
flat_x = np.asarray(x).flatten()
```

```
return multivariate_normal.logpdf(flat_x, flat_mean,
                                   cov, allow_singular=allow_singular)
```

```
def fx_ballistic(x, dt):
```

```
    """
```

9D ballistic model (CV-CA-IMM): state = [x, vx, ax, y, vy, ay, z, vz, az].

Constant-acceleration discretisation:

```
    p_new = p + v*dt + 0.5*a_total*dt^2
```

```
    v_new = v + a_total*dt
```

```
    a_new = a (deterministic — it only changes through the Q[a] noise).
```

```
    """
```

```
    vx = np.clip(x[1], -40.0, 40.0)
```

```
    vy = np.clip(x[4], -40.0, 40.0)
```

```
    vz = np.clip(x[7], -40.0, 40.0)
```

```
    ax_s, ay_s, az_s = x[2], x[5], x[8]
```

```
    k = 0.0314
```

```
    v_mag = np.sqrt(vx**2 + vy**2 + vz**2)
```

```
    a_drag_x = -k * v_mag * vx
```

```
    a_drag_y = -k * v_mag * vy
```

```
    a_drag_z = -k * v_mag * vz
```

```
    g = 9.81
```

```
    a_tot_x = ax_s + a_drag_x
```

```
a_tot_y = ay_s + a_drag_y - g
```

```
a_tot_z = az_s + a_drag_z
```

```
return np.array([  
    x[0] + vx*dt + 0.5*a_tot_x*dt**2,  
    vx + a_tot_x*dt,  
    ax_s,  
    x[3] + vy*dt + 0.5*a_tot_y*dt**2,  
    vy + a_tot_y*dt,  
    ay_s,  
    x[6] + vz*dt + 0.5*a_tot_z*dt**2,  
    vz + a_tot_z*dt,  
    az_s,  
])
```

```
def fx_hit(x, dt):
```

```
    """9D hit model: identical dynamics to fx_ballistic. The hit mode differs  
    only in a much larger Q[a] (set in create_imm_estimator), which lets the  
    acceleration wake from ~0 to impulsive values within a frame or two."""  
    return fx_ballistic(x, dt)
```

```
def fx_bounce(x, dt):
```

```
    """
```

```
    9D floor-bounce model: vy is inverted with epsilon=0.75, vx/vz are damped  
    with friction=0.85.
```

```
    """
```

|     |      |          |       |      |                    |       |
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```

epsilon = 0.75
friction = 0.85
x_next = np.copy(x)
vx_new = x[1] * friction
vy_new = -x[4] * epsilon
vz_new = x[7] * friction

x_next[0] += vx_new * dt
x_next[3] += vy_new * dt
x_next[6] += vz_new * dt

x_next[1] = vx_new
x_next[4] = vy_new
x_next[7] = vz_new

# Reset residual acceleration — a bounce is a "fresh start".
x_next[2] = 0.0
x_next[5] = 0.0
x_next[8] = 0.0

return x_next

```

```
def hx(x):
```

```
    """
```

Measurement function 9D -> 3D: extract positions only (indices 0, 3, 6).  
 Velocities (1, 4, 7) and accelerations (2, 5, 8) are not measured, only  
 estimated by the UKF/IMM.



''''''

```
return np.array([x[0], x[3], x[6]])
```

```
def measurement_transform(prediction_data, K, R_matrix, camera_pos,  
                           ball_diameter=0.21):
```

''''''

Turn a detection record (px, py, bbox width) into a 3D world measurement [X, Y, Z] in court coordinates, with the convention:

- X — across the court (along the end line),
- Y — vertical axis (up, perpendicular to the floor),
- Z — along the long side (along the side line).

''''''

```
u_cam, v_cam, w_cam = (prediction_data['x_pos'], prediction_data['y_pos'],  
                        prediction_data['w_box'])
```

```
raw_x, raw_y, raw_z = get_3d_position(u_cam, v_cam, w_cam, K,  
                                       R_matrix, camera_pos, ball_diameter)
```

```
measurement = np.array([raw_x, raw_y, raw_z], dtype=np.float32)
```

```
return measurement
```

```
def get_dynamic_transition_matrix(y, vy,  
                                   mahalanobis_sq=0.0,  
                                   hit_residual_min_sq=8.0,  
                                   hit_residual_max_sq=11.34,  
                                   bounce_height_max=0.55,  
                                   frames_since_update=0,  
                                   bounce_max_coast=3,  
                                   m_hit_target=None):
```

"""Return the 3x3 IMM Markov transition matrix for [ballistic, hit, bounce] for the next prediction step. The base matrix sticks to ballistic (M[0,0]=0.95); two triggers steer it: a passive geometric bounce trigger and a residual-based mid-air hit trigger interpolated over [hit\_residual\_min\_sq, hit\_residual\_max\_sq] (the chi^2\_3 95th percentile to the gating bound), clamped outside that range.

y, vy: ball vertical position (m) and velocity (m/s) in world frame.

mahalanobis\_sq: squared Mahalanobis distance of the last accepted detection (0 for tracks without updates).

"""

```
M = np.array([[0.95, 0.04, 0.01],
              [0.60, 0.40, 0.00],
              [0.90, 0.00, 0.10]])
```

```
if frames_since_update > bounce_max_coast:
```

```
    return np.eye(3)
```

```
if (y < bounce_height_max and vy < 0 and frames_since_update
    <= bounce_max_coast):
```

```
    M[0] = [0.10, 0.05, 0.85]
```

```
    M[1] = [0.10, 0.05, 0.85]
```

```
    return M
```

```
if mahalanobis_sq > hit_residual_min_sq:
```

```
    alpha = min(
```

```
        1.0,
```

```
        (mahalanobis_sq - hit_residual_min_sq)
```

```

        / max(hit_residual_max_sq - hit_residual_min_sq, 1e-6),
    )
    M_hit_target = (np.array([0.45, 0.50, 0.05])
                    if m_hit_target is None
                    else np.asarray(m_hit_target, dtype=float))
    M[0] = (1.0 - alpha) * M[0] + alpha * M_hit_target

    return M

```

### **ball\_3d\_analysis/IMMTracker.py**

```

import numpy as np
from scipy.optimize import linear_sum_assignment
from IMM_UKF import *
import cv2

```

```

def create_imm_estimator(z_initial, dt=0.02, v_initial=None):
    """Build an IMM estimator from three UKFs (ballistic / hit / bounce).

    z_initial: world position [x, y, z] at init.
    dt:        time step (= 1 / FPS).
    v_initial: optional initial velocity [vx, vy, vz] in m/s. None -> zeros.

    Passing (z_k - z_{k-1}) / dt speeds up UKF convergence and
    avoids residual/Mahalanobis spikes in a new track's first frames.
    """

    # 9D constant-acceleration state: [x, vx, ax, y, vy, ay, z, vz, az].
    dim_x = 9

```

```

dim_z = 3
points = MerweScaledSigmaPoints(n=dim_x, alpha=.1, beta=2., kappa=1.)
# Initial covariance. Accel variance ~25 (m/s^2)^2 lets the filter absorb
# the first non-zero acceleration without immediately blowing the gate;
# velocity variance 50 covers the finite-difference bootstrap spread.
P_init = np.diag([
    0.1, 50.0, 25.0,
    0.1, 50.0, 25.0,
    0.1, 50.0, 25.0,
])
# Measurement noise (m^2). sigma_z ~ 0.7 m reflects monocular depth error
# (Z_c = f*D / bbox_w); tighter values made trajectories too jagged.
R_init = np.diag([0.02, 0.02, 0.5])

# Process noise per model. In Q_discrete_white_noise(dim=3) var is the
# jerk variance (bottom-right block cell), i.e. how much accel may drift.
# ballistic: small jerk noise (free flight is smooth, residual accel ~0).
q_var_ballistic = 0.1
q_b = Q_discrete_white_noise(dim=3, dt=dt, var=q_var_ballistic)
Q_ballistic = block_diag(q_b, q_b, q_b)

# hit: large jerk noise lets accel jump from ~0 to tens of m/s^2 in a frame
# or two. The impulse is modelled through accel growth (Singer-style CA),
# not a raw velocity jump that broke gating in the old 6D version.
q_var_hit = 400
q_h = Q_discrete_white_noise(dim=3, dt=dt, var=q_var_hit)
Q_hit = block_diag(q_h, q_h, q_h)

```

```

# bounce: moderate jerk noise; fx_bounce already resets accel to 0 on
# contact, Q[a] only guards against abrupt deviations in later frames.
q_var_bounce = 5
q_bnc = Q_discrete_white_noise(dim=3, dt=dt, var=q_var_bounce)
Q_bounce = block_diag(q_bnc, q_bnc, q_bnc)

# Ballistic filter
ukf_ballistic = UnscentedKalmanFilter(name='ballistic UKF', dim_x=dim_x,
                                       dim_z=dim_z, dt=dt, fx=fx_ballistic, hx=hx, points=points)
ukf_ballistic.P = P_init
ukf_ballistic.Q = Q_ballistic
ukf_ballistic.R = R_init

# Hit filter
ukf_hit = UnscentedKalmanFilter(name='hit UKF', dim_x=dim_x, dim_z=dim_z,
                                dt=dt, fx=fx_hit, hx=hx, points=points)
ukf_hit.P = P_init
ukf_hit.Q = Q_hit
ukf_hit.R = R_init

# Bounce filter
ukf_bounce = UnscentedKalmanFilter(name='bounce UKF', dim_x=dim_x,
                                   dim_z=dim_z, dt=dt, fx=fx_bounce, hx=hx, points=points)
ukf_bounce.P = P_init
ukf_bounce.Q = Q_bounce
ukf_bounce.R = R_init

filters_lt = [ukf_ballistic, ukf_hit, ukf_bounce]

```

```

mu = np.array([0.95, 0.04, 0.01])
M_base = np.array([[0.95, 0.04, 0.01],
                    [0.60, 0.40, 0.00],
                    [0.90, 0.00, 0.10]])

imm = IMMEstimator(filters_lt, mu, M_base)

# Initialise the 9D state. Use v_initial if given, else zero velocity.
# Accel always starts at 0 (ballistic prior); the high P_init[a] lets the
# filter discover a non-zero acceleration when needed.
if v_initial is None:
    vx0 = vy0 = vz0 = 0.0
else:
    vx0, vy0, vz0 = (float(v_initial[0]), float(v_initial[1]),
                     float(v_initial[2]))
initial_x = np.array([
    z_initial[0], vx0, 0.0,
    z_initial[1], vy0, 0.0,
    z_initial[2], vz0, 0.0,
])
for f in imm.filters:
    f.x = initial_x.copy()
    # IMMEstimator copies f.x_post into imm.x_post once, in its constructor,
    # before we overwrite f.x. Without re-setting f.x_post here, the birth
    # snapshot logs zeros -> overlay draws a (0,0,0) world point. x_post is
    # only a logging snapshot, so this does not affect predict/update.
    f.x_post = initial_x.copy()
imm._compute_state_estimate()

```

# Same for the IMM aggregate read by snapshot\_track in run\_segment.

imm.x\_post = initial\_x.copy()

return imm

class Track:

# Sanity cap on bootstrap velocity (m/s): above this the finite-difference

# almost certainly linked two different objects, so bootstrap is skipped.

bootstrap\_max\_speed = 35.0

# Track-level covariance reset on a radical residual. A high Mahalanobis<sup>2</sup>

# (~5% tail of  $\chi^2_3$ ) means an impulsive manoeuvre (hit/block); we

# "open up" velocity/accel covariance before imm.update(z) so the new

# measurement pulls the state onto the new trajectory without multi-frame lag.

cov\_reset\_mah\_sq\_threshold = 8.0 # ~95th percentile of  $\chi^2_3$

cov\_reset\_inflate\_v = 50.0 # (m/s)<sup>2</sup> — same level as P\_init[v]

cov\_reset\_inflate\_a = 100.0 # (m/s<sup>2</sup>)<sup>2</sup> — 4x P\_init[a]

# Delayed-accept hysteresis. A suspicious detection (mah\_sq > threshold) is

# frozen in \_pending and waits for confirmation from later frames: if the

# next detection is also far, the manoeuvre is real; if it returns to

# normal, the pending was a false positive and is dropped.

hysteresis\_mah\_sq\_threshold = 8.0 # suspicion threshold (= cov\_reset)

hysteresis\_confirm\_threshold = 5.0 # next mah\_sq below this -> pending was FP

hysteresis\_window = 6 # max frames to wait for confirmation

coast\_threshold = 4 # consecutive misses = coasting: high

# mah\_sq is expected (P grew naturally),

# not a manoeuvre -> no inflate, no defer

```
# Coast covariance cap. While coasting P grows unbounded; large P_vel
# spreads the UKF sigma points so wide that the quadratic drag (Jensen)
# pulls the sigma-mean velocity toward zero and overrides gravity -> vy
# freezes and the track "hangs" instead of following a ballistic arc. The
# cap keeps sigma points tight so gravity dominates again. pos cap = 9 m^2
# (+-3 m) matches Gate A so the Mahalanobis gate stays meaningful.
coast_p_pos_cap = 9.0 # (m^2) — +-3 m, = Gate A max_assoc_residual
coast_p_vel_cap = 50.0 # (m/s)^2 — = P_init[v]
coast_p_acc_cap = 100.0 # (m/s^2)^2 — = cov_reset_inflate_a
```

```
def __init__(self, track_id, z_initial, dt, v_initial=None,
              initial_hits=1, min_hits=3, enable_hysteresis=False,
              enable_cov_reset=False, enable_coast_cov_cap=False,
              enable_coast_z_fence=False, coast_z_min=-3.0, coast_z_max=21.0,
              hit_residual_min_sq=8.0, hit_residual_max_sq=11.34,
              bounce_height_max=0.55, bounce_max_coast=3,
              m_hit_target=None):
```

```
"""
```

enable\_cov\_reset: inflate P[v,a] when mah\_sq > threshold. Default False  
(lets the residual live 1-2 frames so the hit trigger  
fires and IMM likelihood raises mu\_hit; cov-reset and  
hit are substitutes for the same residual signal).

enable\_hysteresis: delayed-accept defer of suspicious detections. Default  
False (defer cannibalises cov-reset and the hit trigger  
— all three share mah\_sq=8 — and zeroes the residual  
the next predict needs). Coast-skip via \_missed\_streak  
stays active regardless: that handles occlusion, not defer.



v\_initial: optional initial velocity vector; lets the UKFs start with a real velocity instead of zeros.

initial\_hits: hits the track starts with. Bootstrapped tracks use 2 (two consecutive detections were effectively seen), promoting to Confirmed one frame sooner.

min\_hits: updates needed to reach 'Confirmed'. Forwarded from IMMTracker (CLI) so the swept parameter affects the track state machine.

~~~~~

```
self.track_id = track_id
```

```
self.dt = float(dt)
```

```
self.enable_hysteresis = bool(enable_hysteresis)
```

```
self.enable_cov_reset = bool(enable_cov_reset)
```

```
self.enable_coast_cov_cap = bool(enable_coast_cov_cap)
```

```
# Coast Z-fence: while coasting there is no measurement, so no gate acts  
# and predict() extrapolates world depth Z (idx 6) from a contaminated  
# depth velocity vz (idx 7). Near the frame edge w_box collapses, Z=f*D/w  
# inflates, and coasting carries the ball to Z=33-42 m (court is 18 m).  
# The fence clamps predicted Z to [coast_z_min, coast_z_max] and damps  
# vz/az that push past it. Active only during coasting (mark_missed).
```

```
self.enable_coast_z_fence = bool(enable_coast_z_fence)
```

```
self.coast_z_min = float(coast_z_min)
```

```
self.coast_z_max = float(coast_z_max)
```

```
# IMM triggers, forwarded to get_dynamic_transition_matrix on each  
# predict(). Defaults match the function signature (A/B compatibility).
```

```
self.hit_residual_min_sq = float(hit_residual_min_sq)
```

```
self.hit_residual_max_sq = float(hit_residual_max_sq)
```

```
self.bounce_height_max = float(bounce_height_max)
```

```

self.bounce_max_coast = int(bounce_max_coast)
self.m_hit_target = (None if m_hit_target is None
                      else np.asarray(m_hit_target, dtype=float))
self.imm = create_imm_estimator(z_initial, dt, v_initial=v_initial)

# Remember the first detection for intra-track velocity bootstrap on the
# second hit. Skip it if v_initial was already supplied externally.
self.z_initial = np.asarray(z_initial, dtype=float).copy()
self._velocity_bootstrapped = (v_initial is not None)

# Lifecycle
self.state = 'Tentative' # 'Tentative', 'Confirmed', 'Deleted'
self.time_since_update = 0
self.hits = int(initial_hits)
self.hit_streak = int(initial_hits)
self.min_hits = int(min_hits)
# Last accepted Mahalanobis^2 (gating residual), passed to
# get_dynamic_transition_matrix to activate the hit trigger on the next
# predict(). Reset to 0 after frames without updates.
self._last_mahalanobis_sq = 0.0
# Hysteresis state
self._pending = None # Optional[(z, mah_sq)]
self._pending_frames_elapsed = 0 # frames since pending was set
self._missed_streak = 0 # consecutive mark_missed (coast detect)
# Position-extraction matrix: indices 0, 3, 6 of the 9D state. Velocities
# and accelerations are not measured, only estimated by the UKF.
self.h_matrix = np.array([
    [1, 0, 0, 0, 0, 0, 0, 0, 0], # x

```

```

[0, 0, 0, 1, 0, 0, 0, 0, 0], # y
[0, 0, 0, 0, 0, 0, 0, 1, 0, 0], # z
], dtype=float)

```

```
def _inflate_covariance(self):
```

```

"""Inflate P[v,a] for every sub-filter and the IMM aggregate."""

```

```

for f in self.imm.filters:

```

```

    f.P[1, 1] += self.cov_reset_inflate_v

```

```

    f.P[4, 4] += self.cov_reset_inflate_v

```

```

    f.P[7, 7] += self.cov_reset_inflate_v

```

```

    f.P[2, 2] += self.cov_reset_inflate_a

```

```

    f.P[5, 5] += self.cov_reset_inflate_a

```

```

    f.P[8, 8] += self.cov_reset_inflate_a

```

```

self.imm.P[1, 1] += self.cov_reset_inflate_v

```

```

self.imm.P[4, 4] += self.cov_reset_inflate_v

```

```

self.imm.P[7, 7] += self.cov_reset_inflate_v

```

```

self.imm.P[2, 2] += self.cov_reset_inflate_a

```

```

self.imm.P[5, 5] += self.cov_reset_inflate_a

```

```

self.imm.P[8, 8] += self.cov_reset_inflate_a

```

```
def _collapse_accel_to_ballistic(self):
```

```

"""Zero the residual acceleration (indices 2, 5, 8) in all states.

```

Called while coasting (mark_missed). Without a measurement, predict() keeps integrating stale accel into velocity ($v_x \leftarrow v_x + a_x \cdot dt$) and the track accelerates across the court. Physically the ball between contacts has zero residual accel (only g + drag, already driving terms in fx_*), so collapsing to ballistic keeps coasting at constant velocity. Velocity,

position and P are left untouched.

"""

```
for f in self.imm.filters:
```

```
    f.x[2] = f.x[5] = f.x[8] = 0.0
```

```
    f.x_post[2] = f.x_post[5] = f.x_post[8] = 0.0
```

```
self.imm.x[2] = self.imm.x[5] = self.imm.x[8] = 0.0
```

```
self.imm.x_post[2] = self.imm.x_post[5] = self.imm.x_post[8] = 0.0
```

```
@staticmethod
```

```
def _cap_P_matrix(P, caps):
```

```
    """PD-safe cap of the P diagonal to caps[i]. For each i with
    P[i,i] > caps[i] scale row and column i by s = sqrt(cap/P[i,i]). This
    congruence transform D*P*D preserves positive-definiteness (unlike a
    naive P[i,i]=cap, which can break PD and fail the UKF Cholesky); the
    off-diagonals scale proportionally, keeping correlations valid."""
```

```
    for i in range(P.shape[0]):
```

```
        if P[i, i] > caps[i]:
```

```
            s = np.sqrt(caps[i] / P[i, i])
```

```
            P[i, :] *= s
```

```
            P[:, i] *= s
```

```
def _cap_coast_covariance(self):
```

```
    """Cap P-diagonal growth while coasting (see coast_p_*_cap). Without it
    P grows unbounded, the UKF sigma points spread, the quadratic drag damps
    gravity in the sigma-mean, vy freezes and the track hangs instead of
    following a ballistic arc. Complements Gate A."""
```

```
    caps = np.array([
```

```
        self.coast_p_pos_cap, self.coast_p_vel_cap, self.coast_p_acc_cap,
```

```

        self.coast_p_pos_cap, self.coast_p_vel_cap, self.coast_p_acc_cap,
        self.coast_p_pos_cap, self.coast_p_vel_cap, self.coast_p_acc_cap,
    ])

```

```

for f in self.imm.filters:

```

```

    self._cap_P_matrix(f.P, caps)

```

```

self._cap_P_matrix(self.imm.P, caps)

```

```

def _fence_coast_depth(self):

```

```

    """Clamp world depth Z (idx 6) while coasting to [coast_z_min,
    coast_z_max] and damp the depth velocity/accel (vz=7, az=8) that push
    past the fence. Applied to all UKF filters (their x and x_prior seed the
    next sigma points) and to the mixed imm state (reported in overlay).

```

```

    P is left to Gate C (coast_cov_cap)."""

```

```

    zlo, zhi = self.coast_z_min, self.coast_z_max

```

```

def _clamp(x):

```

```

    if x[6] > zhi:

```

```

        x[6] = zhi

```

```

        if x[7] > 0.0:

```

```

            x[7] = 0.0

```

```

            x[8] = 0.0

```

```

        elif x[6] < zlo:

```

```

            x[6] = zlo

```

```

            if x[7] < 0.0:

```

```

                x[7] = 0.0

```

```

                x[8] = 0.0

```

```

for f in self.imm.filters:

```

```

    _clamp(f.x)
    _clamp(f.x_prior)
    _clamp(self.imm.x)
    _clamp(self.imm.x_prior)

```

```
def predict(self):
```

```
    # 9D state: y = x[3], vy = x[4].
```

```

    self.imm.M = get_dynamic_transition_matrix(
        self.imm.x[3], self.imm.x[4],
        mahalanobis_sq=self._last_mahalanobis_sq,
        frames_since_update=self.time_since_update,
        hit_residual_min_sq=self.hit_residual_min_sq,
        hit_residual_max_sq=self.hit_residual_max_sq,
        bounce_height_max=self.bounce_height_max,
        bounce_max_coast=self.bounce_max_coast,
        m_hit_target=self.m_hit_target,
    )

```

```
    self.imm.predict()
```

```
    if self._pending is not None:
```

```
        self._pending_frames_elapsed += 1
```

```
        if self._pending_frames_elapsed > self.hysteresis_window:
```

```
            # Confirmation window exhausted: drop pending, resume ageing.
```

```
            self._pending = None
```

```
            self._pending_frames_elapsed = 0
```

```
            self.time_since_update += 1
```

```
        # else: track does not age during an active hysteresis window
```

```
    else:
```

```
        self.time_since_update += 1
```

```

def update(self, z, R=None):
    # R: optional per-detection measurement covariance (adaptive sigma_Z).
    # None -> the IMM filters use their own fixed self.R.
    current_mah_sq = self._last_mahalanobis_sq

    # Delayed-accept hysteresis. Defer (pending) runs only with
    # enable_hysteresis. Coast-skip (via _missed_streak) is always active:
    # it handles return from occlusion, not detection deferral.
    _coast_skip = False # True -> skip inflate and the hysteresis trigger

    if self.enable_hysteresis and self._pending is not None:
        pending_z, _ = self._pending
        self._pending = None
        self._pending_frames_elapsed = 0

    if current_mah_sq >= self.hysteresis_confirm_threshold:
        # Both frames far from the prediction -> real manoeuvre. Inflate
        # P and update with the current z. pending_z (the late
        # measurement) is not applied: a second imm.update() without a
        # predict() between them breaks PD of P (Cholesky fail).
        self._inflate_covariance()
        self.imm.update(z, R=R)
        self.time_since_update = 0
        self.hits += 1
        self.hit_streak += 1
        if self.state == 'Tentative' and self.hits >= self.min_hits:
            self.state = 'Confirmed'

```

```

        self._missed_streak = 0
        return

    # mah_sq < confirm threshold -> the detection returned to a normal
    # trajectory -> pending was a false positive. Continue with a normal
    # update, no inflate and no new trigger.
    _coast_skip = True
    self._missed_streak = 0

    elif self._missed_streak >= self.coast_threshold:
        # Long miss streak -> high mah_sq is expected (P grew naturally while
        # coasting), not a manoeuvre. Skip inflate and the hysteresis trigger.
        _coast_skip = True
        self._missed_streak = 0

    elif (self.enable_hysteresis and z is not None
          and current_mah_sq > self.hysteresis_mah_sq_threshold):
        # Suspicious detection: defer it for confirmation.
        self._pending = (np.asarray(z, dtype=float).copy(), current_mah_sq)
        self._pending_frames_elapsed = 0
        self.imm.update(None) # extrapolate instead of accepting
        self.hit_streak = 0
        self._last_mahalanobis_sq = 0.0
        self._missed_streak += 1
        return

    # Mechanism A — intra-track velocity bootstrap.
    # On the hits=1 -> 2 transition the UKFs still had v=0 in the prior,

```



```
# which inflates the residual/Mahalanobis and makes the Kalman gain pull
# velocity to truth slowly. We compute v_bootstrap from the finite
# difference (z - z_initial)/(gap*dt) and write it into every filter
# before imm.update, so predict/update run with a correct velocity prior
# from the very first use.
```

```
do_bootstrap = (
```

```
    z is not None
```

```
    and self.hits == 1
```

```
    and not self._velocity_bootstrapped
```

```
    and self.time_since_update > 0
```

```
)
```

```
if do_bootstrap:
```

```
    gap_dt = self.time_since_update * self.dt
```

```
    z_arr = np.asarray(z, dtype=float)
```

```
    v_boot = (z_arr - self.z_initial) / gap_dt
```

```
    speed = float(np.linalg.norm(v_boot))
```

```
    if 0.0 < speed <= self.bootstrap_max_speed:
```

```
        # Write velocity into all sub-filters and their x_prior (which
```

```
        # seeds the next predict sigma points). Accel (2, 5, 8) is left
```

```
        # at 0 — there is no basis to bootstrap it. Velocity indices are
```

```
        # [1, 4, 7] in the 9D state.
```

```
        for f in self.imm.filters:
```

```
            f.x[1] = v_boot[0]
```

```
            f.x[4] = v_boot[1]
```

```
            f.x[7] = v_boot[2]
```

```
            f.x_prior[1] = v_boot[0]
```

```
            f.x_prior[4] = v_boot[1]
```

```
            f.x_prior[7] = v_boot[2]
```

```

# Update the mixed state too.
self.imm.x[1] = v_boot[0]
self.imm.x[4] = v_boot[1]
self.imm.x[7] = v_boot[2]
self.imm.x_prior[1] = v_boot[0]
self.imm.x_prior[4] = v_boot[1]
self.imm.x_prior[7] = v_boot[2]
self._velocity_bootstrapped = True

# Step 2.A — track-level covariance reset on a radical residual. Runs
# after the bootstrap (the bootstrap frame is not a hit) but before
# imm.update(z). _coast_skip means the high mah_sq is expected, so no
# inflate is needed.
do_cov_reset = (
    self.enable_cov_reset
    and z is not None
    and not _coast_skip
    and self._last_mahalanobis_sq > self.cov_reset_mah_sq_threshold
)
if do_cov_reset:
    self._inflate_covariance()

self.imm.update(z, R=R)
self.time_since_update = 0
self.hits += 1
self.hit_streak += 1
if self.state == 'Tentative' and self.hits >= self.min_hits:
    self.state = 'Confirmed'

```

```

self._missed_streak = 0

def mark_missed(self):
    self.hit_streak = 0
    # No measurement -> residual undefined. Reset to 0 so the next predict()
    # does not fire the hit trigger from stale data.
    self._last_mahalanobis_sq = 0.0
    self._missed_streak += 1
    self.imm.update(None)    # extrapolate without a measurement
    # Damp residual accel so coasting does not accelerate by integrating
    # stale ax into velocity.
    self._collapse_accel_to_ballistic()
    # Gate C: cap P-diagonal growth while coasting (keeps extrapolation
    # physical past the frame edge).
    if self.enable_coast_cov_cap:
        self._cap_coast_covariance()
    # Coast Z-fence: keep extrapolated depth physical (<= court size).
    if self.enable_coast_z_fence:
        self._fence_coast_depth()

def get_mahalanobis_distance(self, z, R_matrix):
    """Mahalanobis distance between the IMM prediction and a new detection."""
    # Project the mixed prior state x_prior and covariance P_prior into
    # measurement space.
    z_mean = np.dot(self.h_matrix, self.imm.x_prior)
    S = np.dot(self.h_matrix, np.dot(self.imm.P_prior,
                                     self.h_matrix.T)) + R_matrix

```

```
y = z - z_mean # innovation
```

```
try:
```

```
    S_inv = np.linalg.inv(S)
```

```
    dist_sq = np.dot(y.T, np.dot(S_inv, y))
```

```
    return dist_sq
```

```
except np.linalg.LinAlgError:
```

```
    return 1e5
```

```
class IMMTracker:
```

```
    def __init__(self, dt=0.02, max_age=80, min_hits=3, gating_threshold=16.0,
        bootstrap_max_gap_frames=5, bootstrap_max_dist=2.5,
        bootstrap_max_speed=35.0, enable_hysteresis=False,
        enable_cov_reset=False, enable_single_ball_nms=True,
        enable_physical_gate=True, max_assoc_residual=3.0,
        enable_coast_cov_cap=False, spawn_suppress_max_coast=0,
        enable_depth_robust_gate=False, depth_hold_gain=0.5,
        depth_innov_free=0.0,
        enable_coast_z_fence=False, coast_z_min=-3.0, coast_z_max=21.0,
        tentative_max_age=5, floor_kill_y=-0.3,
        enable_adaptive_depth_R=False,
        hit_residual_min_sq=8.0, hit_residual_max_sq=11.34,
        bounce_height_max=0.55, bounce_max_coast=3,
        m_hit_target=None):
```

```
    """
```

```
    dt:            time step (= 1 / FPS).
```

```
    max_age:       frames without an update before a track is deleted.
```

min_hits: frames to confirm a track (Tentative -> Confirmed).

gating_threshold: Mahalanobis χ^2 gate. $\chi^2_3 p=0.99 = 11.34$;
 16.0 admits borderline manoeuvre detections
 (mah 12-15) to revive the hit mode without growth.

bootstrap_max_gap_frames: max gap (frames) between a pending detection
 and the current one to bootstrap velocity. Default 5.

bootstrap_max_dist: max 3D distance (m) between pending and current
 detection for bootstrap. Default 2.5 m.

bootstrap_max_speed: sanity cap (m/s); above it bootstrap is skipped
 (likely a false link of two objects). Default 35.

enable_hysteresis: delayed-accept hysteresis per Track. Default False
 (defer cannibalises cov-reset + hit).

enable_cov_reset: Step 2.A covariance reset per Track. Default False
 (Path B) to keep the IMM hit mode alive.

enable_single_ball_nms: one ball -> at most one Confirmed track. Extra
 confirmed tracks (phantom duplicates) are removed,
 keeping the best. Default True.

enable_physical_gate: Gate A. A covariance-independent ceiling on the
 Euclidean residual $\|z - \text{prediction}\|$. The Mahalanobis
 gate depends on P, which inflates while coasting and
 lets an 8 m FP jump pass; a fixed ceiling does not.
 Default True.

max_assoc_residual: Gate A threshold (m). Real-manoevre residuals reach
 ~ 2.2 m; the nearest teleport jumps to ~ 6 m, so 3.0 m
 cleanly separates them.

~~~~~

self.dt = dt

self.max\_age = max\_age

```

self.min_hits = min_hits
self.gating_threshold = gating_threshold

self.bootstrap_max_gap_frames = int(bootstrap_max_gap_frames)
self.bootstrap_max_dist = float(bootstrap_max_dist)
self.bootstrap_max_speed = float(bootstrap_max_speed)
self.enable_hysteresis = bool(enable_hysteresis)
self.enable_cov_reset = bool(enable_cov_reset)
self.enable_single_ball_nms = bool(enable_single_ball_nms)
# Gate A: covariance-independent physical ceiling on the residual.
self.enable_physical_gate = bool(enable_physical_gate)
self.max_assoc_residual = float(max_assoc_residual)
# Gate C: cap P-diagonal growth while coasting (forwarded to Track).
self.enable_coast_cov_cap = bool(enable_coast_cov_cap)
# Gate D: suppress spawning a parasite track while the real Confirmed
# track is briefly coasting (0 < tsu <= spawn_suppress_max_coast). One
# ball domain: a jumpy occluded detection used to spawn a competitor that
# then stole identity. 0 = disabled (A/B compatibility); ~8 covers a
# typical hand occlusion at 50 FPS.
self.spawn_suppress_max_coast = int(spawn_suppress_max_coast)
# Gate A-depth: depth-robust Gate A. Split the residual into lateral
# (X, Y — the pixel projection YOLO gives correctly even under occlusion)
# and depth (Z — monocular f*D/w, which degrades as w collapses). Gate
# only the lateral part hard (hypot(dX,dY) <= max_assoc_residual) and,
# instead of rejecting a large depth innovation, strongly distrust it by
# inflating sigma_Z^2. Depth is then held by the prediction while X,Y
# update from the valid detection. Default False (A/B compatibility).
self.enable_depth_robust_gate = bool(enable_depth_robust_gate)

```

```

self.depth_hold_gain = float(depth_hold_gain)

# Huber-style dead zone for depth distrust: damp only the excess of
# |innov| over a normal-physics threshold. excess = max(0, |innov| -
# depth_innov_free); sigma_Z^2 += (gain*excess)^2. Pure (gain*innov)^2
# also damped legitimate depth (far side, serve); ~3 m dead zone passes
# normal physics and suppresses only occlusion outliers (6-11 m).
# 0.0 = disabled. Recommended ~3.0 (= max_assoc_residual).
self.depth_innov_free = float(depth_innov_free)

# Coast Z-fence: while coasting no gate acts and predict() extrapolates
# world Z from a contaminated vz (w_box collapses near the frame edge ->
# Z=f*D/w inflates -> vz up to +/-19 m/s, Z carried to 33-42 m on an 18 m
# court). The fence in each track's mark_missed clamps predicted Z to
# [coast_z_min, coast_z_max] and damps vz/az. Default False. Recommended
# coast_z_max~21 (back line 18 m + ~3 m for servers), coast_z_min~-3.
self.enable_coast_z_fence = bool(enable_coast_z_fence)

self.coast_z_min = float(coast_z_min)
self.coast_z_max = float(coast_z_max)

# Defect 1: ghost tracks. A Tentative track that never reached min_hits
# must not coast as a full track — 1-2 noise detections otherwise draw
# an ~80-frame fiction. Kill Tentative tracks after this many coast frames.
self.tentative_max_age = int(tentative_max_age)

# Defect 2: sub-floor coasting. A ball in flight cannot be below the
# floor; if a coasting track extrapolates y < floor_kill_y, terminate it.
# Small negative margin against floor noise (ball r ~0.1 m).
self.floor_kill_y = float(floor_kill_y)

# Adaptive sigma_Z. If True and detection_covs is passed to update(),
# gating and the IMM update use per-detection 3x3 R (anisotropic, along
# the ray; sigma_Zc = f*D/w^2 * sigma_w) instead of the fixed

```

```

# self.R_matrix. Default False (A/B compatibility).
self.enable_adaptive_depth_R = bool(enable_adaptive_depth_R)
# IMM triggers, forwarded to every Track -> get_dynamic_transition_matrix.
# Allow tuning bounce/hit activation from the CLI. Defaults match the
# function signature (A/B compatibility).
self.hit_residual_min_sq = float(hit_residual_min_sq)
self.hit_residual_max_sq = float(hit_residual_max_sq)
self.bounce_height_max = float(bounce_height_max)
self.bounce_max_coast = int(bounce_max_coast)
self.m_hit_target = (None if m_hit_target is None
                      else np.asarray(m_hit_target, dtype=float))
# Gate B: merge-identity guard. Two Confirmed tracks inherit a shared id
# only if physically reconcilable (same ball split by a manoeuvre, within
# this distance). A distant FP track that reached Confirmed does not
# donate its id to the real coasting track (that would amplify teleports).
# Manoeuvre fragments stay <=~2 m, FP teleports >=6 m, so 4.0 m separates.
self.single_ball_merge_max_dist = 4.0
# Coasting <= this many frames does not cost identity (we bucket tsu).
# tol=1: a real track survives one missed frame (detection noise) but
# after 2+ coast frames yields to an actively-detected track. A small tol
# minimises the spatial jump at a switch.
self.single_ball_nms_tsu_tol = 1

self.tracks = []
self.next_id = 1
# Matrix for Mahalanobis gating, consistent with R_init in
# create_imm_estimator (sigma_z ~ 0.7 m for monocular depth).
self.R_matrix = np.diag([0.02, 0.02, 0.5])

```



```

# Ring buffer of unassigned detections from the last few frames. Used to
# bootstrap velocity when spawning a new track: if the current
# unassigned detection is close to one from the last N frames, init the
# track with  $v = (z\_now - z\_prev)/(dframe*dt)$  instead of zeros.
self._spawn_buffer = [] # elements: (np.array z, int frame_idx)
self._frame_counter = 0

```

```

def update(self, detections_3d, detection_covs=None):

```

```

    """

```

```

    detections_3d: list of np.array([x, y, z]) for all detected objects
    in the frame.

```

```

    detection_covs: optional parallel list of 3x3 measurement covariances
    (one per detection). Used only if enable_adaptive_depth_R=True;
    None -> fixed self.R_matrix.

```

```

    """

```

```

    self._frame_counter += 1

```

```

    def _R_for(d_idx):

```

```

        """R for detection d: adaptive if enabled and provided."""

```

```

        if (self.enable_adaptive_depth_R
            and detection_covs is not None
            and detection_covs[d_idx] is not None):

```

```

            return detection_covs[d_idx]

```

```

        return self.R_matrix

```

```

    def _depth_robust_R(track, z, R_base):

```

```

        """Inflate  $\sigma_Z^2$  proportional to the squared depth innovation so

```

a large depth jump barely moves the filter (depth held by the prediction, X,Y updated from the valid detection). z\_pred from x\_prior (predict already ran for all tracks)."""

```
z_pred = np.dot(track.h_matrix, track.imm.x_prior)
```

```
depth_innov = float(z[2] - z_pred[2])
```

```
# Dead zone: damp only the excess of |innov| over normal physics.
```

```
excess = max(0.0, abs(depth_innov) - self.depth_innov_free)
```

```
R_eff = np.array(R_base, dtype=float, copy=True)
```

```
R_eff[2, 2] += (self.depth_hold_gain * excess) ** 2
```

```
return R_eff
```

```
for track in self.tracks:
```

```
    track.predict()
```

```
if len(detections_3d) == 0:
```

```
    for track in self.tracks:
```

```
        track.mark_missed()
```

```
    self._age_spawn_buffer()
```

```
    self._manage_lifecycle()
```

```
    return
```

```
if len(self.tracks) == 0:
```

```
    # No tracks yet. Any detection may bootstrap if the spawn buffer has
```

```
    # a predecessor.
```

```
    for z in detections_3d:
```

```
        self._spawn_track_from_unmatched(z)
```

```
    self._age_spawn_buffer()
```

```
    return
```

```

cost_matrix = np.full((len(self.tracks), len(detections_3d)), 1e5)

for t, track in enumerate(self.tracks):
    # Gate A reference point: the track's predicted position h*x_prior.
    z_pred = np.dot(track.h_matrix, track.imm.x_prior)
    for d, z in enumerate(detections_3d):
        R_d = _R_for(d)
        # Gate A: reject a pair whose Euclidean jump from the prediction
        # is physically impossible in one step, independent of how much P
        # inflated while coasting (that is what opened the Mahalanobis
        # gate for teleports).
        if self.enable_physical_gate:
            if self.enable_depth_robust_gate:
                # Gate A-depth: gate only the lateral (X,Y) part. The
                # large depth innovation is not rejected but handled by
                # depth distrust (inflated sigma_Z in gating and update),
                # so a detection with valid u,v is accepted.
                lateral = float(np.hypot(z[0] - z_pred[0],
                                          z[1] - z_pred[1]))
                if lateral > self.max_assoc_residual:
                    continue

                R_d = _depth_robust_R(track, z, R_d)
            elif np.linalg.norm(z - z_pred) > self.max_assoc_residual:
                continue

            dist_sq = track.get_mahalanobis_distance(z, R_d)
            if dist_sq < self.gating_threshold:
                cost_matrix[t, d] = dist_sq

```

```

# Association (Hungarian algorithm).
row_ind, col_ind = linear_sum_assignment(cost_matrix)

unmatched_tracks = set(range(len(self.tracks)))
unmatched_detections = set(range(len(detections_3d)))

# Update matched pairs.
for r, c in zip(row_ind, col_ind):
    if cost_matrix[r, c] < self.gating_threshold:
        # Save Mahalanobis^2 at accept time; it feeds
        # get_dynamic_transition_matrix on the next predict().
        self.tracks[r]._last_mahalanobis_sq = float(
            cost_matrix[r, c]
        )
        R_c = _R_for(c)
        if self.enable_depth_robust_gate:
            # Same inflated sigma_Z as in gating: depth held by the
            # prediction, X,Y updated from the detection.
            R_c = _depth_robust_R(self.tracks[r],
                                   detections_3d[c], R_c)
        self.tracks[r].update(detections_3d[c], R=R_c)
        unmatched_tracks.remove(r)
        unmatched_detections.remove(c)

# Missed tracks.
for t in unmatched_tracks:
    self.tracks[t].mark_missed()

```

```

# Spawn new tracks for unmatched detections (with velocity bootstrap if a
# predecessor is found). Gate D: if a real Confirmed track is briefly
# coasting ( $0 < tsu \leq \text{spawn\_suppress\_max\_coast}$ ), do not spawn a
# competitor from outlier detections — and keep the outlier out of the
# spawn buffer so a later legitimate spawn starts with zero velocity.
suppress_spawn = (
    self.spawn_suppress_max_coast > 0
    and any(
        t.state == 'Confirmed'
        and  $0 < t.time\_since\_update \leq \text{self.spawn\_suppress\_max\_coast}$ 
        for t in self.tracks
    )
)
if not suppress_spawn:
    for d in unmatched_detections:
        self._spawn_track_from_unmatched(detections_3d[d])

# Age the spawn buffer and remove stale tracks.
self._age_spawn_buffer()
self._manage_lifecycle()

# -----
# Initial-velocity bootstrap helpers
# -----

def _spawn_track_from_unmatched(self, z_now):
    """Create a new Track for the unassigned detection z_now. If the spawn
    buffer holds a predecessor that satisfies the constraints (close in

```

space, recent in time, consistent velocity), init the track with  
 $v\_initial = (z\_now - z\_prev)/(dframe*dt)$  and hits=2; otherwise a plain  
 zero-velocity init. The used spawn point is removed (one-to-one) to avoid  
 gluing two tracks to the same historical detection.””””

```
z_now_arr = np.asarray(z_now, dtype=float)
```

```
best_idx = -1
```

```
best_dist = self.bootstrap_max_dist
```

```
best_v = None
```

```
best_gap = 0
```

```
for i, (z_prev, f_prev) in enumerate(self._spawn_buffer):
```

```
    gap = self._frame_counter - f_prev
```

```
    if gap <= 0 or gap > self.bootstrap_max_gap_frames:
```

```
        continue
```

```
    d_xyz = float(np.linalg.norm(z_now_arr - z_prev))
```

```
    if d_xyz >= best_dist:
```

```
        continue
```

```
    gap_dt = gap * self.dt
```

```
    if gap_dt <= 0:
```

```
        continue
```

```
    v_cand = (z_now_arr - z_prev) / gap_dt
```

```
    speed = float(np.linalg.norm(v_cand))
```

```
    if speed > self.bootstrap_max_speed:
```

```
        # Velocity beyond physically possible -> no bootstrap.
```

```
        continue
```

```
    best_idx = i
```

```
    best_dist = d_xyz
```

```
    best_v = v_cand
```

```

best_gap = gap

if best_idx >= 0:
    # Bootstrap: create a track with non-zero v and hits=2.
    self.tracks.append(
        Track(self.next_id, z_now_arr, self.dt,
              v_initial=best_v, initial_hits=2,
              min_hits=self.min_hits,
              enable_hysteresis=self.enable_hysteresis,
              enable_cov_reset=self.enable_cov_reset,
              enable_coast_cov_cap=self.enable_coast_cov_cap,
              enable_coast_z_fence=self.enable_coast_z_fence,
              coast_z_min=self.coast_z_min,
              coast_z_max=self.coast_z_max,
              hit_residual_min_sq=self.hit_residual_min_sq,
              hit_residual_max_sq=self.hit_residual_max_sq,
              bounce_height_max=self.bounce_height_max,
              bounce_max_coast=self.bounce_max_coast,
              m_hit_target=self.m_hit_target)
    )
    self.next_id += 1
    # Remove the used spawn point.
    del self._spawn_buffer[best_idx]
else:
    # Plain zero-velocity spawn + buffer the detection for a future
    # bootstrap.
    self.tracks.append(
        Track(self.next_id, z_now_arr, self.dt,

```

```

        min_hits=self.min_hits,
        enable_hysteresis=self.enable_hysteresis,
        enable_cov_reset=self.enable_cov_reset,
        enable_coast_cov_cap=self.enable_coast_cov_cap,
        enable_coast_z_fence=self.enable_coast_z_fence,
        coast_z_min=self.coast_z_min,
        coast_z_max=self.coast_z_max,
        hit_residual_min_sq=self.hit_residual_min_sq,
        hit_residual_max_sq=self.hit_residual_max_sq,
        bounce_height_max=self.bounce_height_max,
        bounce_max_coast=self.bounce_max_coast,
        m_hit_target=self.m_hit_target)
    )
    self.next_id += 1
    self._spawn_buffer.append(
        (z_now_arr.copy(), self._frame_counter)
    )

```

```
def _age_spawn_buffer(self):
```

```
    """Drop buffer entries older than bootstrap_max_gap_frames."""
```

```
    cutoff = self._frame_counter - self.bootstrap_max_gap_frames
```

```
    self._spawn_buffer = [
```

```
        (z, f) for (z, f) in self._spawn_buffer if f > cutoff
```

```
    ]
```

```
def _manage_lifecycle(self):
```

```
    active_tracks = []
```

```
    for track in self.tracks:
```



```

# Delete a track that has not updated for too long.
if track.time_since_update > self.max_age:
    track.state = 'Deleted'

# Defect 1: ghost track. Tentative (never confirmed) but already
# coasting past tentative_max_age frames -> noise spawn. Confirmed
# tracks are untouched.
elif (track.state == 'Tentative'
      and track.time_since_update > self.tentative_max_age):
    track.state = 'Deleted'

# Defect 2: sub-floor coast. Only when the track has no measurement
# (coasting): with a detection y is measured and no clamp is needed.
elif (track.time_since_update > 0
      and track.imm.x[3] < self.floor_kill_y):
    track.state = 'Deleted'

# Keep the track unless deleted.
if track.state != 'Deleted':
    active_tracks.append(track)

self.tracks = active_tracks

if self.enable_single_ball_nms:
    self._suppress_duplicate_confirmed()

def _suppress_duplicate_confirmed(self):
    """Single-ball NMS. One ball in play -> at most one Confirmed track. If

```

several exist (phantom duplicates from noise spawns or divergent coasting), keep the best and DELETE the rest.

Deletion (not demotion) is deliberate: a demoted duplicate keeps stealing detections in the Hungarian step, re-confirms and competes again, producing identity switches. Deletion sends detections to the winner; if the winner truly diverges, a fresh detection spawns a new track (slightly higher fragmentation, much smoother trajectory: jerk down, switches 9->3).

Best = (min bucketed-tsu, max hits, min last Mahalanobis<sup>2</sup>): actively updated > long-lived > well-fitted.

tsu bucketing: a short coast ( $tsu \leq tsu\_tol$ ) counts as 0. Without it one missed frame on a real track ( $tsu=1$ ) would hand identity to a divergent ghost with  $tsu=0$ , causing identity flicker during manoeuvres. With the bucket the real track keeps identity through short turbulence; a ghost with a LONG coast ( $tsu > tsu\_tol$ ) still yields to the active track. """

```
confirmed = [t for t in self.tracks if t.state == 'Confirmed']
```

```
if len(confirmed) < 2:
```

```
    return
```

```
def _key(t):
```

```
    eff_tsu = 0 if t.time_since_update <= self.single_ball_nms_tsu_tol \
```

```
        else t.time_since_update
```

```
    return (eff_tsu, -t.hits, t._last_mahalanobis_sq)
```

```
confirmed.sort(key=_key)
```

```
winner = confirmed[0]
```

```

# Merge-identity (fragmentation fix): one ball domain, so all coexisting
# Confirmed tracks are the same physical ball split by a manoeuvre. The
# winner inherits the minimum id of the group so logical identity is
# continuous across the seam. Tracks split by a real trajectory break
# (different rallies) never coexist as Confirmed, so no false merge.
# Carry the max hits too, so the merged track stays a stable NMS winner.
#
# Gate B (merge guard): inherit an id only from physically reconcilable
# members (<= single_ball_merge_max_dist from the winner). A distant FP
# track that reached Confirmed does not donate its id (that would
# legalise a teleport); it is simply deleted as a duplicate.
def _pos(t):
    x = t.imm.x_post
    return np.array([x[0], x[3], x[6]])

wp = _pos(winner)
reconcilable = [
    t for t in confirmed
    if np.linalg.norm(_pos(t) - wp) <= self.single_ball_merge_max_dist
]
winner.track_id = min(t.track_id for t in reconcilable)
winner.hits = max(t.hits for t in reconcilable)

for loser in confirmed[1:]:
    loser.state = 'Deleted'
self.tracks = [t for t in self.tracks if t.state != 'Deleted']

```

```
def get_confirmed_tracks(self):
```

```
    """Return only the tracks we are confident in (Confirmed state)."""
```

```
    return [t for t in self.tracks if t.state == 'Confirmed']
```

### **training\_models/model\_train\_remote.py**

```
import os
```

```
import torch as t
```

```
import torch.nn as nn
```

```
import torch.nn.functional as F
```

```
from pathlib import Path
```

```
from torchvision import transforms
```

```
from torchvision.transforms import Compose, ToTensor, PILToTensor, Resize
```

```
from tqdm import tqdm
```

```
import numpy as np
```

```
# import optuna as opt
```

```
from ultralytics import YOLO
```

```
import albumentations as A
```

```
main_model = YOLO('../parsing_datasets_scripts/yolo26m.pt')
```

```
custom_transforms = [
```

```
    A.MotionBlur(blur_limit=(7, 15), p=0.4),
```

```
    A.GaussNoise(std_range=(0.01, 0.05), p=0.2),
```

```
    A.CLAHE(clip_limit=4.0, p=0.3),
```

```
    A.RandomBrightnessContrast(brightness_limit=0.25,
```

```
                               contrast_limit=0.25, p=0.4),
```

```
    A.HueSaturationValue(hue_shift_limit=15, sat_shift_limit=30,
```

```

        val_shift_limit=20, p=0.3),
    ]

# Purely for server training
main_model.train(data='/workspace/diploma_project/data/server_data.yaml',
                  epochs=100, device='0', imgsz=640,
                  augmentations=custom_transforms, patience=20)

metrics = main_model.val(data='/workspace/diploma_project/data'
                          '/server_data.yaml', imgsz=640)

print(f'mAP50: {metrics.box.map50}')
print(f'mAP50-95: {metrics.box.map}')

```